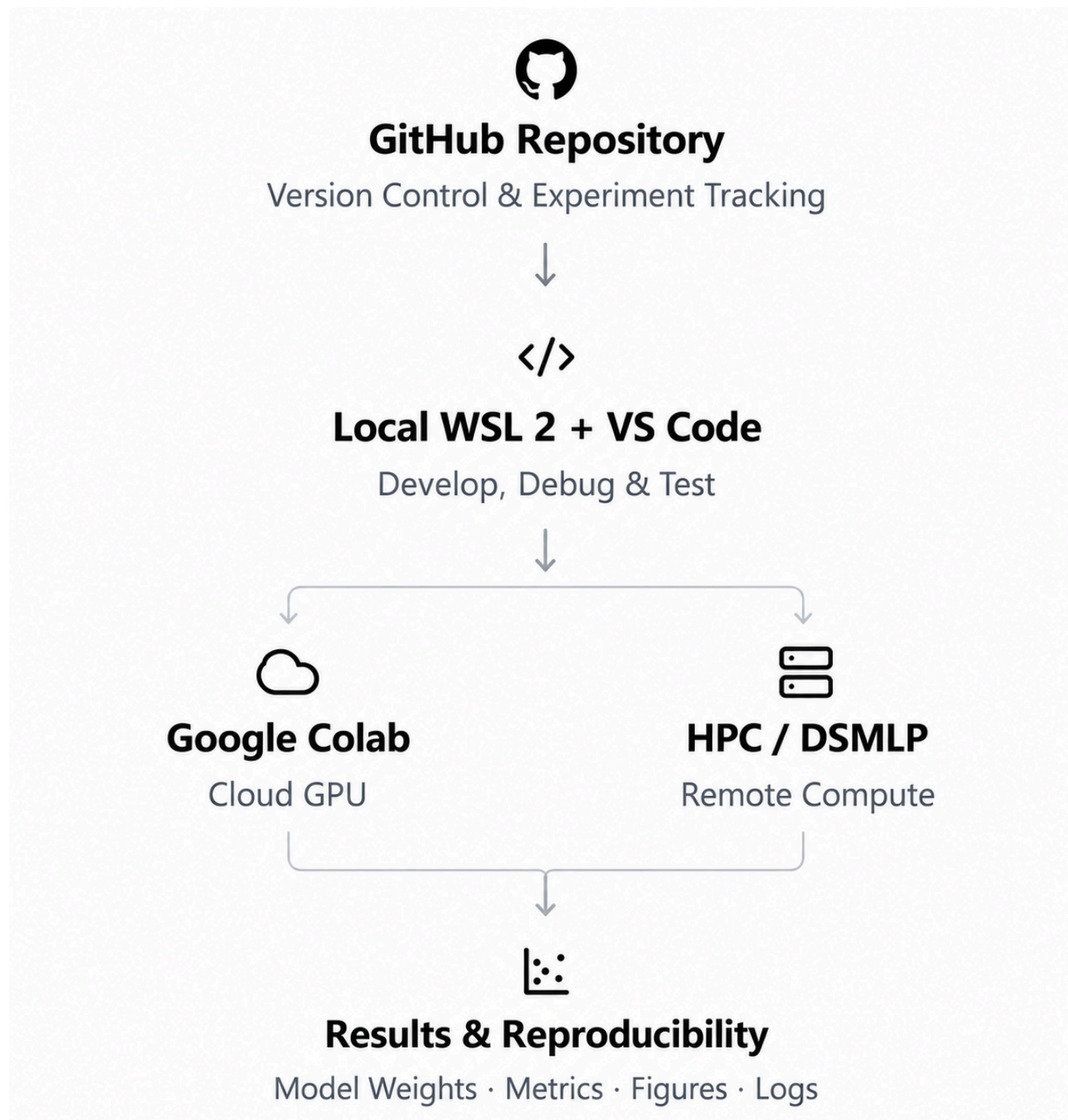


Introduction

Research Workflow



A Microsoft technology that uses lightweight virtualization to run a real Linux kernel and Linux environment directly within Windows, without requiring a dual-boot setup

Environment:

1.Ubuntu

[Get Ubuntu](#) | [Download](#) | [Ubuntu](#)

An opensource operation system based on Linux

Suitable for Python/PyTorch/Scientific Research Computing

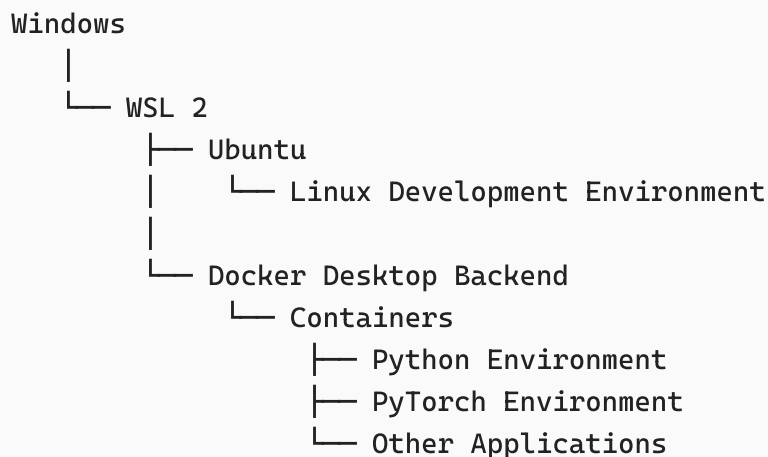
2.Docker

[Docker](#)

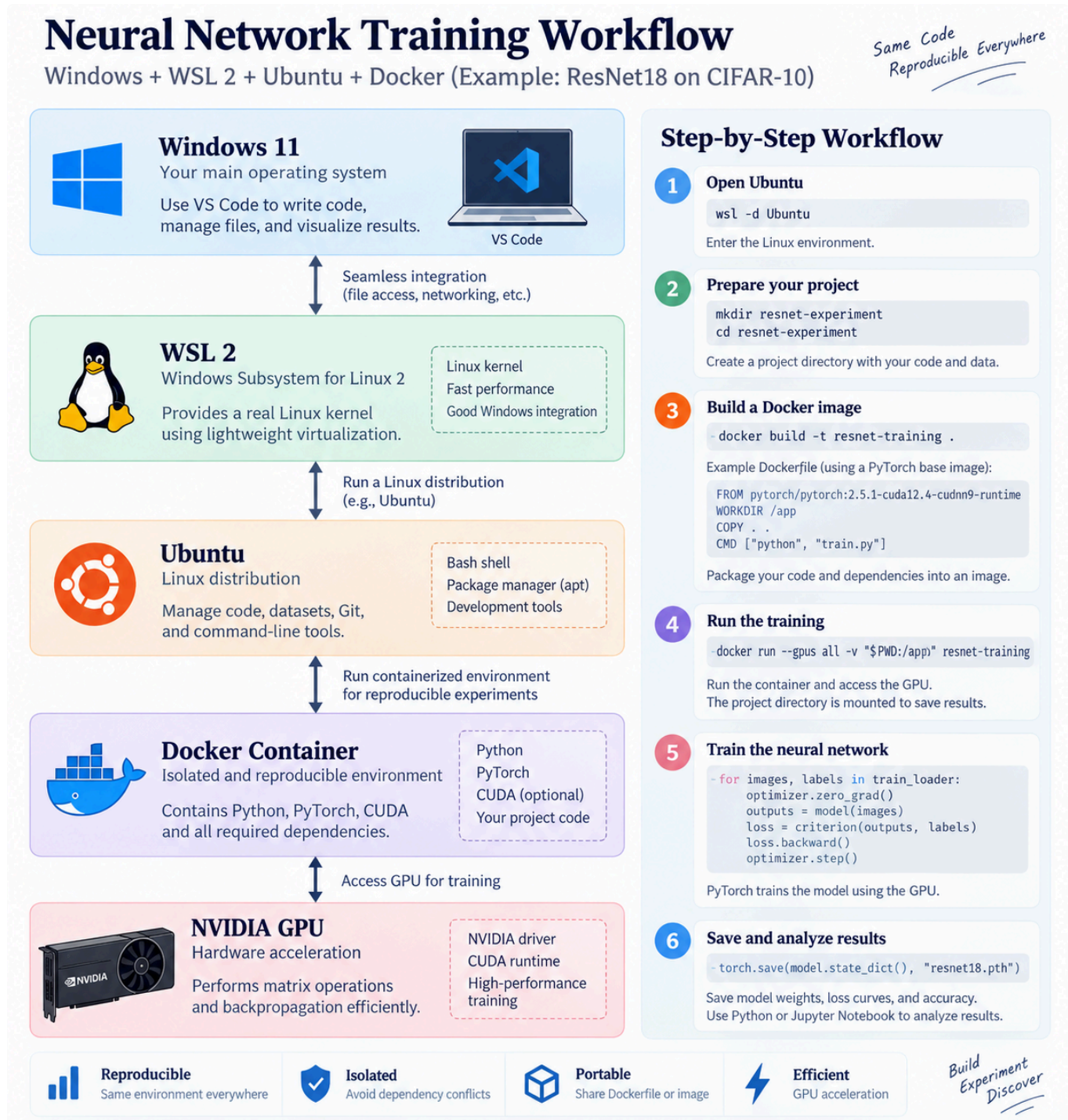
[Docker](#) | [Secure Sandboxes for AI Agents](#)

Docker is a containerization platform that enables developers to build, deploy, and run applications in isolated, portable, and reproducible environments called containers.

relationships:



classical workflow:

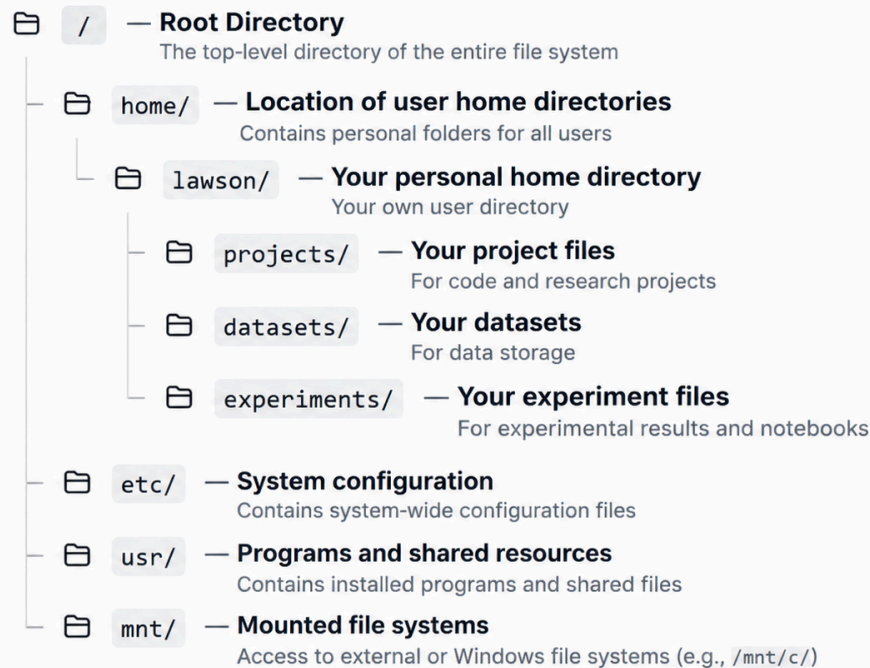


pwd: Print Working Directory
ls: List
cd: Change Directory

Linux Home Directory

Linux Home Directory is a user's personal directory in the Linux file system, used to store files, projects, and user-specific configurations.

2. Linux File System Structure



APT

APT (Advanced Package Tool) is a package management tool used to install, update, upgrade, and remove software packages on Debian-based Linux distributions such as Ubuntu.



Ubuntu Terminal

```
sudo apt install python3
```



APT Package Manager

Resolves dependencies, downloads and installs software packages



Software Repositories

Ubuntu's configured software repositories

Python Virtual Environment Tool

A Python virtual environment tool (such as `venv`) creates isolated Python environments for individual projects, allowing them to manage dependencies independently and avoid package version conflicts.

Suppose you are working on two neural network research projects:

- **Project A:** using PyTorch 2.5.
- **Project B:** using PyTorch 2.8.

If the two projects share the same Python environment, installing different versions of PyTorch may cause conflicts.

With virtual environments:

